

ACTIVITY 11

LEVELS:

Grade 12, University Preparation

COURSES:

Canada and World Issues
HEV.02, HEV.03, UCV.01,
HE1.01, HE2.02, HE2.04, GI1.02,
GI2.02, GI2.06, GI2.07, GI3.01

TASK:

Through GIS processes,
students investigate the current
and changing nature of agricultural
resources and sustainability.

EXPECTATIONS:

The student will:

- Analyse the impact of past and current trends in agriculture on natural and human systems.
- Demonstrate an understanding of the technologies used in the analysis and synthesis of geographic data.
- Produce mental, thematic, and choropleth maps to illustrate local or global patterns.

TIMEFRAME:

90 Minutes

ACHIEVEMENT CATEGORIES:

Knowledge / Understanding
Thinking / Inquiry
Communication
Application / Making Connections

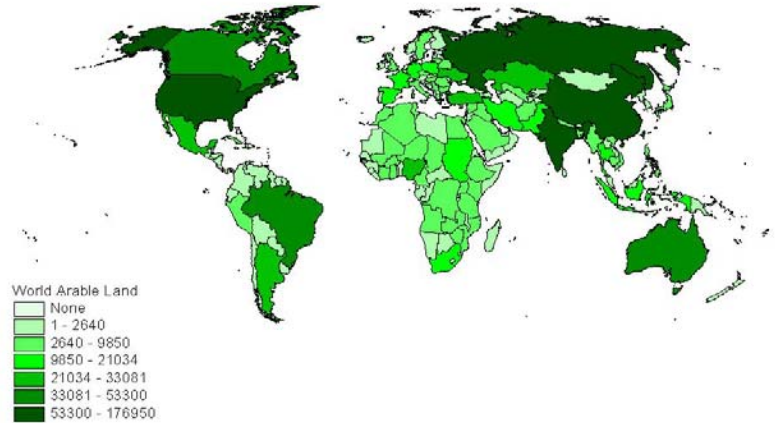
MATERIALS:

Computer system with ArcView
GIS 3.2 software
Worksheet 12
Cartographic Skills and Rubric

VOCABULARY:

Renewable resources
Environmental carrying
Capacity
Per capita
Sustainability

Sq. Km. of Arable Land



Global Sustainable Agriculture

PROCEDURE

1. Students will research background information on global agricultural resources and sustainable agriculture.
2. Students will download data sets needed for GIS analysis of global agriculture (exports, imports, labour, production, fertilizer etc.). The data can be found on the website <http://www.ocdsb.edu.on.ca/woodsweb/Geomatics> and downloaded before doing the lesson.
3. Students will follow the defined GIS exercise to develop products that provide insight into the current and changing global agriculture.
4. Students will use the GIS output products to answer questions that are embedded into the exercise.

DISCUSSION: AGRICULTURE OF THE WORLD

KNOWLEDGE / UNDERSTANDING

1. What is sustainable agriculture? Compare agriculture in several different countries (developed, developing, undeveloped) in terms of arable land, production, imports and exports, and fertilizer use.
2. How can countries move towards sustainable agriculture?
3. Which countries show positive change and which countries show negative change in terms of sustainable agriculture? Why?

THINKING / INQUIRY

Start a discussion of the following:

4. What do we need to know in order to measure sustainable agriculture?
5. What is needed to improve global sustainable agriculture (e.g. leadership, communication, trust, legislation, technology, trade, understanding of consequences, examples of failure, etc.)?
6. How does GIS analysis help to understand the complexity of global sustainable agriculture?

HOMEWORK SUGGESTION

Research paper:

1. Select an additional aspect of agriculture. (E.g. Livestock, tractors, rice, etc.)
2. Research the selected topic for 10 different countries to support a background paper and a GIS project.
3. Find, select and acquire data for the 10 countries to be used in the GIS project. Add the data to the world.dbf for GIS products.
(http://earthtrends.wri.org/searchable_db/index.cfm?theme=10)
4. Produce GIS products (maps, charts or tables) to support the background paper.
5. Define the implications for sustainable development in the 10 countries.